

```
    yo.t = 1;

    if (yo.y || yo.t)

        printf("Admit");

    else if (yo.y && yo.t)

        printf("Not Admit");

    else

        printf("Clear");

    return 0;

}
```

☐ Nothing will be displayed


```
int main()
{
    union yoo
    {
        int y;
        int t;
    }yo;
    yo.y = 2;
    yo.t = 1;
    if (yo.y || yo.t)
        printf("Admit");
    else if (yo.y && yo.t)
        printf("Not Admit");
```


50 #include <stdio.h>

int main()

51 {

#define amk(a, b) a || b

int a = 4, b = 6;

a = amk(a - 1, b + 1);

b = a + amk(b - 1, a + 1);

printf("%d", amk(a, b));

return 0;

}


```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    #define amk(a, b) a || b
```

```
    int a = 4, b = 6;
```

```
    a = amk(a - 1, b + 1);
```

```
    b = a + amk(b - 1, a + 1);
```

```
    printf("%d", amk(a, b));
```

```
    return 0;
```

```
}
```


What will be the output of the program given below?

```
#include <stdio.h>

static int _1_, _1_;

int main()
{
    _1_ = _1_ = 2;

    _1_ = 3;

    printf("%d %d ", _1_, _1_);

    return 0;
}
```



```
#include <stdio.h>

#define pull 45 //line 2

#define pull1 46 //line 3

int main()
{
    int zx;

    zx = pull1; //line 7

    pull1 = pull; //line 8

    pull = zx; //line 9

    printf("%d\n%d", pull, pull1); //line 10

    return 0;

}
```

Find the erroneous line numbers.


```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    #define amk(a, b) a || b
```

```
    int a = 4, b = 6;
```

```
    a = amk(a - 1, b + 1);
```

```
    b = a + amk(b - 1, a + 1);
```

```
    printf("%d", amk(a, b));
```

```
    return 0;
```

```
}
```


30

```
int main()
```

31

```
{
```

```
    char *s="fun(10,20) "
```

32 ☆

```
    "s = fun(*s,10)";
```

```
    printf(s);
```

33

```
    return 0;
```



34

```
}
```

```
int fun(int a, int b)
```

35

```
{
```

```
    int *c;
```

36

```
    return (*c = (a+b));
```

37

```
}
```


Point out the error if any in the program given below?

```
#include <stdio.h>

int main()
{
    struct a
    {
        signed var1 : 5;           //Line 6
        signed var2 : 4;           //Line 7
    };

    printf("%d ", sizeof(struct a)); //Line 9
    printf("%d ", sizeof(struct a obj)); //Line 10
    return 0;
}
```



```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char *s="fun(10,20) "
```

```
    "s = fun(*s,10)";
```

```
    printf(s);
```

```
    return 0;
```

```
}
```

```
int fun(int a, int b)
```

```
{
```

```
    int *c;
```

```
    return (*c = (a+b));
```

```
}
```



```
int main()
```

```
{
```

```
    char *s="fun(10,20) "
```

```
    "s = fun(*s,10)";
```

```
    printf(s);
```

```
    return 0;
```

```
}
```

```
int fun(int a, int b)
```

```
{
```

```
    int *c;
```

```
    return (*c = (a+b));
```

```
}
```


Which of the below given is/are general register(s)?

I) EAX

II) ESI

III) EIP

☐ Only (I)

☒ All (I), (II) and (III)

☐ Only (II)

☐ Only (I) and (II)

30 #include <stdio.h>

1 int main()

{

2 char *s="fun(10,20) "

3 "s = fun(*s,10)";

printf(s);

return 0;

}

int fun(int a, int b)

{

int *c;

return (*c = (a+b));

}


```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, x;
```

```
    for (i = 5; i <= (10 % 3 + i); i--)
```

```
    {
```

```
        x = (i * 5 - i / 3) * i;
```

```
        printf("%d", x);
```

```
    }
```

```
    return 0;
```

```
}
```

☐ Run time error

22 #include <stdio.h>
23 #ifndef ij
24 #define s(i, j) 111j
25 ☆ #define mn 111
int main()
{
char m = '1', n = '2';
printf("%d\n", mn);
#if (mn == 111)
printf("%d ", s(m, n));
#undef mn
#endif
#endif
return 0;
}


```
#include <stdio.h>
#ifdef i
#define s(i, j) 1111
#define mn 111
int main()
{
    char m = '1', n = '2';
    printf("%d\n", mn);
    #if (mn == 111)
    printf("%d ", s(m, n));
    #undef mn
    #endif
    #endif
    return 0;
```

```
}
```


49

```
#include <stdio.h>
#include <string.h>
int main()
{
    char s[20], *p;
    printf("Enter a string");
    scanf("%s", s);
    p=s;
    while (*p)
    {
        if(*p >= 97 && *p<= 122)
            *p = *p - 32;
        p++;
    }
    puts(s);
    return 0;
```

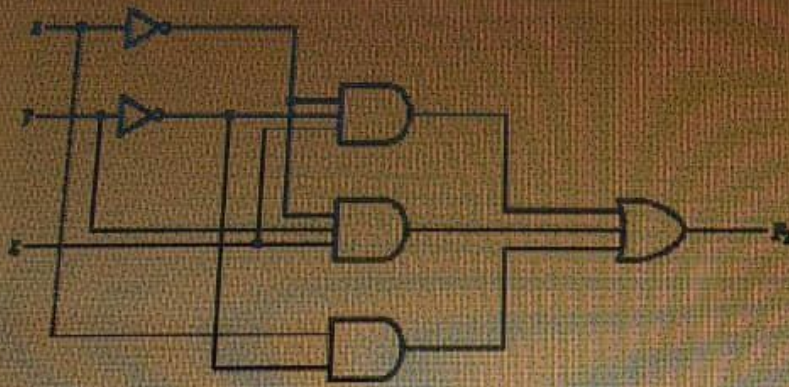
51

2

3

}

Which one of the options is the final equation after simplifying boolean function for the circuit given below



☐ $F_2 = X'Z + XY'$

☐ $F_2 = XZ' + X'Y$

☐ $F_2 = X'Y + XZ'$

☐ $F_2 = X'Y'Z + X'YZ + XY'$

Previous Question

Next Question


```
23 int main()
24 {
25     union yo;
26     {
27         int y;
28         int t;
29     }yo;
30     yo.y = 2;
31     yo.t = 1;
32     if (yo.y || yo.t)
        printf("Admit");
    else if (yo.y && yo.t)
        printf("Not Admit");
    else
        printf("Clear");
    return 0;
}
```


#include <stdio.h>

int f(int x)

{

++x;

return x;

}

int main(void)

{

printf("%d", f(3));

return 0;

}

☐ Compile time error

☐ 4

27

Mark(s): 1

Programming Basics

SECTION: C Programming

28

What will be the output of the program given below?

29

```
#include <stdio.h>
```

30

```
#define kap(s) kapp(p) + 2
```

```
#define kapp(p) p && p++ || ++s
```

31

```
int main()
```

32

```
{
```

33

```
    int p = 0x12, s = 0x06, i;
```

```
    s += kap(s);
```

34

```
    printf("%d", s += kapp(p) + p--);
```

```
    return 0;
```

35

```
}
```

36

☐ 14

37

☐ 8

38 ☆

☐ 12

39

☐ 9

40

[Previous Question](#)[Next Question](#)

27

28

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37 ☆

38

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40

QUESTION: 37

GROUP: Programming Basics

SECTION: Data Structures Basics

Mark(s): 1

Consider a full circular queue 'S' of size 6 where FRONT = 0 and REAR = 5. How many Dequeue and Enqueue operations are required to get

☐ Dequeue 3 and Enqueue 3☐ Dequeue 3 and Enqueue 2☐ Dequeue 1 and Enqueue 4☐ Dequeue 2 and Enqueue 3[Previous Question](#)[Next Question](#)

27

28

29

30

31

32 ☆

33

34

35

36

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38

39

40

Which of the line number from the given below program will give a compile time error:

```
#include <stdio.h>

int a;

a = 30, 40; //Line 3

int b = 1, 2; //Line 4

int main()
{
    int c = 3, 4; //Line 8
    int d;
    d = 10, 20; //Line 10
    printf("%d %d %d %d ", a, b, c, d);
    return 0;
}
```

☐

Line 3, line 4

☐

Line 3, line 4, line 8, line 10

☐

Line 3, line 4, line 8

☐

No error

[Previous Question](#)[Next Question](#)

25

26

27

28

29

30 ☆

31

32

33

34

35

36

37

38

39

Instructions

QUESTION: 30

GROUP: Programming Basics

SECTION: Computer Architecture

Mark(s): 1

Given the boolean expression $Y = (A + A.B)' + A$. Which of the given option will be a simplified expression?

☐ 1☐ AB☐ A☐ B

Previous Question

Next Question

What will be the output of the program given below?

```
#include <stdio.h>
int main()
{
    unsigned char i = 0x80 ;
    printf("%d", i << 1);
    return 0;
}
```

☐ 100

☐ 0

☐ 256

☐ Garbage value

39 ☆

40

23 ★

24

25

26

27

28 ☆

29

30

31

32

33

34

35

36

37

QUESTION: 28

Mark(s): 1

GROUP: Programming Basics

SECTION: Operating System Fundamentals

What happens to the average turn around time, when the time quantum increases in case of the Round Robin Scheduling Algorithm?

☐ It Remains constant☐ It Increases☐ It Decreases☐ It Varies regularly

Previous Question

Next Question

23 ★

QUESTION: 27 GROUP: Programming Basics SECTION: C Programming

Mark(s): 1

24

25

26

27 ☆

28

29

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35

36

What one of the given options is correct about the program given below?

```
#include <stdio.h>

int main()
{
    int *ptr = "dfghci";
    int **p = &ptr;
    ****p++;
    printf("%d", --****p);
    return 0;
}
```

- ☐ It will print : g
- ☐ It will show a warning message.
- ☐ It will print : f
- ☐ It will show compile time error.

[Previous Question](#)[Next Question](#)

21

22 ☆

23

24

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34

QUESTION: 22
Mark(s): 1

GROUP: Programming Basics

SECTION: Operating System Fundamentals

Consider the factors given below :

- I) Page size
- II) Instruction set architecture
- III) Physical memory size
- IV) Processes in the memory

Which of the factor(s) decide(s) the least number of page frames that should be allocated to a running process in a system implementing the virtual memory?

- ☐ Only (III)
- ☐ Only (II)
- ☐ Only (II) and (III)
- ☐ Only (I) and (IV)

[Previous Question](#)[Next Question](#)

22

Mark(s): 1

GROUP: Programming Basics

SECTION: Operating System Fundamentals

TIME LEFT

23

Which of the given below statement(s) is/are true regarding segmentation?

24

I) Entire segment may temporarily be copied into an available region of physical memory.

25

II) Segment may be divided into pages which can be individually copied into physical memory.

26

☐ Only (I)

27

☐ Neither (I) nor (II)

28

☐ Both (I) and (II)

29

☐ Only (II)

30

31

32

33

34

[Previous Question](#)[Next Question](#)

Direction: Study the information given below and answer the question that follows:

The information given below is regarding the number of employees hired in three different departments: Sales, Service and Marketing in four different years 2017, 2018, 2019 and 2020.

In 2017

The employees who got hired in the Sales department were twice the employees who got hired for the Service department. The total number of employees hired in three departments together was 2400. The number of employees who got hired in the Service department was 60 more than employees who got hired in the Marketing department.

In 2018

The employees who got hired in the Service department were 40% more than employees who got hired in the Sales department while employees who got hired in the Marketing department were 20% less than employees who got hired in the Service department. The total number of employees hired in all three departments together was 4224.

In 2019

The ratio of employees hired in the Sales department to the Service department was 5:6. employees got hired in the Marketing department was 15% more than employees got hired in Service department. The total number of employees hired in all three departments together was 2506.

In 2020

The average number of employees hired in the Sales and Service departments was 478. The number of employees hired in the Marketing department was 600. The ratio of employees for Sales to that for Marketing was 4:5.

Question:

What is the difference between the number of employees hired in the Service and Marketing department together in 2017 and the number of employees hired in the Sales and Marketing department together in 2019?

Previous Question

Next Question

24

25

QUESTION: 29 GROUP: Programming Basics SECTION: Computer Architecture
Mark(s): 1

26

27

What is the final value generated for A in the below given expression? (All the numbers are in binary)

28

$$A + 1101 = 11011$$

29 ☆

☐ 1110

30

☐ 0100

31

☐ 0111

32

☐ 1010

33

34

35

36

37

38

Previous Question

Next Question

QUESTION: 14 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

The minimum value of the expression $(a^2 + 3a + 1)(b^2 + 3b + 1)(c^2 + 3c + 1)/abc$, where a, b and c are all positive real numbers, is:

☐ 113/23

☐ Cannot be determined

☐ 125

☐ 27

Previous Question

Next Question

QUESTION: 13 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

A student answered 35 problems in a test correctly and was awarded 70%. How many problems were there in the test, if all the problems were worth the same marks?

☐ 35

☐ 50

☐ 85

☐ 25

Previous Question

Next Question

7
8
9
10
11
12 ★
13 ☆
14
15
16
17
18
9

QUESTION: 13 GROUP: Aptitude SECTION: Problem Solving
Mark(s): 1

A student answered 35 problems in a test correctly and was awarded 70%. How many problems were there in the test, if all the problems were worth the same?

☐ 35

☐ 50

☐ 85

☐ 25

Previous Question

Next Question

A sum of Rs. 15000 is distributed among Gauri, Parvati and Lakshmi in the ratio of 6 : 5 : 4 respectively. The difference between the shares of Gauri and Lakshmi is:

☐ Rs. 1000

☐ Rs. 4000

☐ Rs. 2000

☒ Rs. 3000

[Previous Question](#)

[Next Question](#)

QUESTION: 10 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

TIME LEFT 00

'P' bought a land for Rs. 4 Lakhs two years back and decided to sell it at an additional 50% of the cost price as profit. What would be the selling price of the land?

☐ Rs. 600000

☐ Rs. 6000000

☐ Rs. 8000000

☐ Rs. 800000

Previous Question

Next Question

QUESTION: 9 GROUP: Aptitude SECTION: Puzzles

Mark(s): 1

Arupam Bar

TIME LEFT 00:2

Kia started off from her house towards the South to a fair. After walking for 50 m, she took a left turn and walked for another 50 m. Then she took three left turns consecutively walking 10 m between each turn. Then, she realized that she was lost and remembered the actual way. So, she took a right turn and walked for 50 m. Then she took two consecutive right turns walking 10 m between each turn and finally reached the fair. In which direction is the fair from her house what is the shortest distance between her house and the fair?

- ☐ West, 40 m
- ☐ North-West, 30 m
- ☐ North-West, 50 m
- ☐ West, 50 m

Previous Question

Next Question

QUESTION: 8 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

TIME LEFT

'A' rides his bike at a speed of 50 km/hr and covers a distance of 250 km. If he increases his speed by 5 km then what is the distance travelled by him in the same time?

☐ 265 km

☐ 280 km

☐ 275 km

☐ 262 km

Previous Question

Next Question

7
8
9
10
11
12 ★

QUESTION: 20 GROUP: Aptitude SECTION: Problem Solving
Mark(s): 1

A bag contains 5 yellow balloons and 6 blue balloons. What is probability of drawing a blue balloon at random?

☐ 1/5

☐ 1/6

☐ 5/11

☐ 6/11

Previous Question

Next Question

Service department. The total number of employees hired in all three departments together was 2006. employees got hired in the Marketing department was 15% more than employees got hired

In 2020

The average number of employees hired in the Sales and Service departments was 478. The number of employees hired in the Marketing department was 600. The ratio of employees for Sales to that for Marketing was 4:5.

Question:

Find the total number of employees hired in all three departments in all four years together.

☐ 12345

☐ 9865

☐ 10686

☐ 18564

[Previous Question](#)

[Next Question](#)

Ratio of employees hired in the Sales department to the Service department was 5:6, employees got hired in the Marketing department was 15% more than employees got hired in the Service department. The total number of employees hired in all three departments together was 2506.

In 2020

The average number of employees hired in the Sales and Service departments was 478. The number of employees hired in the Marketing department was 600. The ratio of employees hired for Sales to that for Marketing was 4:5.

Question:

The ratio of the number of male to female employees in the Sales department in 2017 was 3:2 and 45% of the total employees hired in the Service department in 2019 were female. What is the sum of the number of female employees hired in the Sales department in 2017 and male employees in the Service department in 2019?

☐ 810

☐ 954

☐ 658

☐ 756

[Previous Question](#)

[Next Question](#)

In 2019

The ratio of employees hired in the Sales department to the Service department was 5:6. employees got hired in the Marketing department was 15% more than employees got hired in the Service department. The total number of employees hired in all three departments together was 4224.

In 2020

The average number of employees hired in the Sales and Service departments was 478. The number of employees hired in the Marketing department was 600. The ratio of employees hired for Sales to that for Marketing was 4:5.

Question:

What is the difference between the number of employees hired in the Service and Marketing department together in 2017 and the number of employees hired in the Sales and Service department together in 2019?

☐ 586

☐ 456

☐ 370

☐ 240

[Previous Question](#)

[Next Question](#)

2 ☆

3

4

5

6

7

8

9

10

11

12

3

The information given below is regarding the number of employees hired in three different departments: Sales, Service and Marketing in four different years 2017, 2018, 2019 and 2020.

In 2017

The employees who got hired in the Sales department were twice the employees who got hired for the Service department. The total number of employees hired in three departments together was 2400. The number of employees who got hired in the Service department was 60 more than employees who got hired in the Marketing department.

In 2018

The employees who got hired in the Service department were 40% more than employees who got hired in the Sales department while employees who got hired in the Marketing department were 20% less than employees who got hired in the Service department. The total number of employees hired in all three departments together was 4224.

In 2019

The ratio of employees hired in the Sales department to the Service department was 5:6. employees got hired in the Marketing department was 15% more than employees who got hired in the Service department. The total number of employees hired in all three departments together was 2506.

In 2020

The average number of employees hired in the Sales and Service departments was 478. The number of employees hired in the Marketing department was 60 more than employees who got hired in the Sales department. The ratio of employees hired in the Sales department to that for Marketing was 4:5.

Question:

What is the difference between the number of employees hired in the Service and Marketing department together in 2017 and the number of employees hired in the Sales and Marketing department together in 2019?

☐ 586

Previous Question

Next Question

1 ☆

2

3

4

5

6

7

8

9

10

11

12

13

14

QUESTION: 1 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

In what proportion, should the butter costing Rs. 510 per kg be mixed with the butter costing Rs. 560 per kg, so that the worth of mixture is Rs. 530 per kg?

☐ 3:2

☐ 5:7

☐ 7:5

☐ 2:3

Previous Question

Next Question

QUESTION: 7 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

A can finish a piece of work in 7 days and B can do the same work in 11 days. Working together, what part of the same work will they be able to complete in one day?

☐ 1/77

☐ 7/11

☐ 11/7

☐ 18/77

Previous Question

Next Question

- 2
- 3
- 4
- 5
- 6
- 7
- 8 ☆
- 9
- 10
- 11
- 12
- 13
- 14
- 15

Direction: Answer the question on the basis of information given below

There are six friends who are interested in further education. So, all of them applied for PhD but in different subjects and in different universities. Some more information is given below about the first name (one is Manish) and last name (one is Kale) of the friends, name of the subject (one is Zoology) in which they applied for PhD and name of the university where they applied.

- I) Sanju is interested in 'Mathematics' and the one whose last name is Pandey applied for PhD in University of Mangalore but not interested in 'Mathematics'.
- II) Harsha applied neither in University of Mangalore nor in University of Karnataka. One whose last name is Jadav applied for PhD in University of Mysore.
- III) Druva applied for PhD in Kuvempu University and he is not interested in 'Criminology' or 'Chemistry'. One whose last name is Pai is interested in 'Anthropology' but he is not Harsha.
- IV) Chandan's last name is not Patil or Pandey and Suresh applied for PhD in University of Hampi. One whose last name is Pai applied for PhD in Kakade University.
- V) One who is interested in Physics applied for PhD in University of Mangalore. Jadav is not interested in 'Criminology'.
- VI) Patil is not the last name of Druva or Sanju. Jadav or Nayak is not last name of Sanju.

Question:

One whose last name is Kale applied for PhD in which of the subject given in options?

- ☐ Zoology
- ☐ Mathematics
- ☐ Anthropology
- ☐ Chemistry

Previous Question

Next Question

QUESTION: 7 GROUP: Aptitude SECTION: Problem Solving
Mark(s): 1

12.5% of the people working for an organization belong to city 'A'. 40% of the remaining people belong to city 'B'. 60% of the remaining people belong to city 'C' and the rest belong to city 'D'. The difference in the number of people belonging to city 'B' and 'D' is 588. What is the total number of people in the organization?

☐ 4800

☐ 4200

☐ 5000

☐ 5600

Previous Question

Next Question

QUESTION: 6 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

A company consists of 15 employees of which 5 are women. If 11 employees are chosen at random to form a team and work on a project, then what is the chance of obtaining at least 3 women in the team?

☐ 11/15

☐ 6/13

☐ 12/13

☐ 7/13

Previous Question

Next Question

QUESTION: 6 GROUP: Aptitude SECTION: Problem Solving

Mark(s): 1

A company consists of 15 employees of which 5 are women. If 11 employees are chosen at random to form a team and work on a project, then what is the chance of obtaining at least 5 women in the team?

☐ 11/15

☐ 6/13

☐ 12/13

☐ 7/13

Previous Question

Next Question

QUESTION: 3 GROUP: Aptitude SECTION: Problem Solving
Mark(s): 1

The present age of Rahul is 4 years more than three times the age of his son. Five years hence Rahul's age will be 15 years more than twice the age of his son. Find

☐ 52 years

☐ 56 years

☐ 48 years

☐ 32 years

Previous Question

Next Question

Mark(a) 1

Instructions

Anupam Bhowik

TIME LEFT 00:15:03

Direction: Answer the question on the basis of information given below.

There are six friends who are interested in further education. So, all of them applied for PhD but in different subjects and in different universities. Some more information is given below about the first name (one is Manish) and last name (one is Kale) of the friends, name of the subject (one is Zoology) in which they applied for PhD and name of the university where they applied.

- I) Sanju is interested in 'Mathematics' and the one whose last name is Pandey applied for PhD in University of Mangalore but not interested in 'Mathematics'.
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- III) Druva applied for PhD in Kuvempu University and he is not interested in 'Criminology' or 'Chemistry'. One whose last name is Pai is interested in 'Anthropology' but he is not Harsha.
- IV) Chandan's last name is not Patil or Pandey and Suresh applied for PhD in University of Hampi. One whose last name is Pai applied for PhD in Kakade University.
- V) One who is interested in Physics applied for PhD in University of Mangalore. Jadav is not interested in 'Criminology'.
- VI) Patil is not the last name of Druva or Sanju. Jadav or Nayak is not last name of Sanju.

Question:

Chandan applied for PhD in which of the University given in options?

☐ University of Karnataka

☐ University of Mangalore

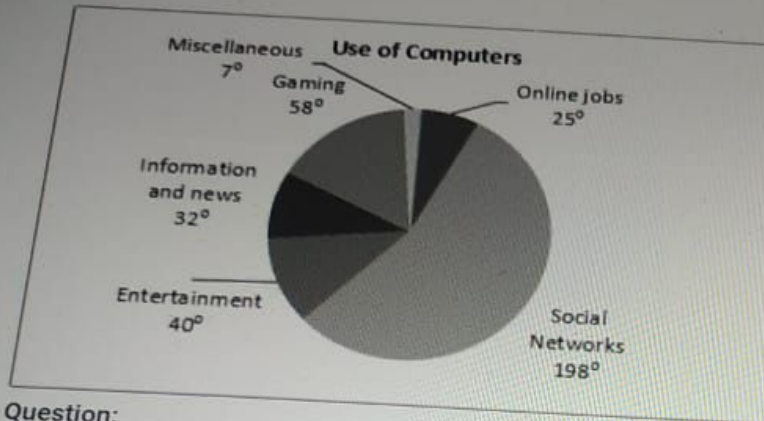
☐ Kakade University

☐ University of Mysore

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2 A survey was done among five thousand adolescent people in a city in India regarding the most preferred use of their personal computers. The data as found from
3 with the help of the following pie diagram:



Question:

How many people approximately use their computers for checking information and news?

☐ 444

☐ 750

☐ 1150

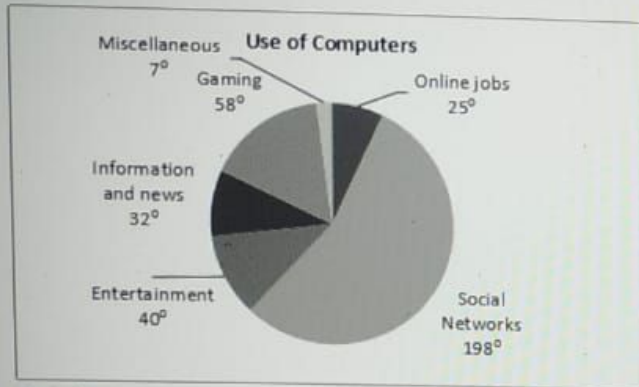
☐ 1600

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Direction: Read the following information and answer the question that follows.

A survey was done among five thousand adolescent people in a city in India regarding the most preferred use of their personal computers. The data as found from the survey is represented with the help of the following pie diagram:



Question:

What is the difference in the number of people who use their computers for entertainment and people who use their computers for online jobs?

☐ 150

☐ 750

☐ 270

☐ 208

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Direction:

P & Q means P is the mother of Q.

P @ Q means P is the brother of Q.

P + Q means P is the son of Q.

P \$ Q means P's wife is Q.

P # Q means P is the daughter of Q.

Then how is 'E' is related to 'D' in the expression of B @ D \$ F # A & C + E?

☐ Mother-in-law

☐ Father-in-law

☐ Brother-in-law

☐ Brother

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47

```
#include <stdio.h>
```

48

```
int main()
```

49

```
{
```

50

```
    printf("size of (void*) = %d\n", sizeof(void*));
```

51

```
    printf("size of (int*) = %d\n", sizeof(int*));
```

52

```
    printf("size of (double*) = %d\n", sizeof(double*));
```

```
    printf("size of (struct unknown*) = %d\n", sizeof(struct unknown *));
```

```
    return 0;
```

```
}
```

53 ☆

54

☒ size of (void*) = 4

size of (int*) = 4

size of (double*) = 4

55

size of (struct unknown*) = 4

56

☐ size of (void*) = 2

size of (int*) = 4

57

size of (double*) = 8

size of (struct unknown*) = 4

58

☐ Compile time error

59

☐ Nothing will be printed

60

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QUESTION: 52 GROUP: Computer Science SECTION: Data Structures
Mark(s): 1

Which of the below given statements are correct about hash tables?

- I) Hash table is an array of fixed size.
- II) Hash functions are used to create hash tables.
- III) Hash table always store keys in ascending order.

☐ Only (I) and (II)

☐ Only (II) and (III)

☐ All (I), (II) and (III)

☐ Only (I) and (III)

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Instructions

47

QUESTION: 51 GROUP: Computer Science SECTION: CA

Mark(s): 1

48

Which of the below given data hazard type(s) is/are also called as Anti -Data Dependency(ies)?

49

I) RAW

50

II) WAR

51 ☆

III) WAW

52

☐ Only (I) and (III)

53

☐ Only (II) and (III)

54

☐ Only (III)

55

☐ Only (II)

56

57

58

59

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```
#include <stdio.h>
```

48

```
int main()
```

```
{
```

49

```
    unsigned int num = 8;
```

50

```
    char *nu = (char *)&num;
```

51

```
    printf("%d ", ~num + function(nu));
```

52

```
    return 0;
```

```
}
```

53

```
int function(char *n)
```

```
{
```

54

```
    *n = *n | *n ;
```

```
    return !(*n);
```

```
}
```

55 ☆

56

☐ -16

57

☐ Compile time error

58

☐ -9

59

☐ -8

60

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QUESTION: 49 GROUP: Computer Science SECTION: OOPS

Mark(s): 1

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47

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Consider the scenario given below:

Two abstract classes A and B have same abstract function test(). Another class C is derived from A and B.

Which of the statements given below are incorrect about the above scenario?

- i) Multiple inheritance is possible only if test() in both the classes share same signature and same return type.
- ii) Multiple inheritance is possible when test() in both the classes share same signature and different return type.
- iii) Only Multi level inheritance is possible as test() is a abstract function.
- iv) Multi level inheritance is possible when test() in both the classes share same signature and different return type.

☐ Only (i), (ii) and (iv)

☐ (i),(ii),(iii) and (iv)

☐ Only (i),(iii) and (iv)

☐ Only (ii),(iii) and (iv)

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QUESTION: 48 GROUP: Computer Science SECTION: Operating System

Mark(s): 1

A counting semaphore has a value of 31. It has 16 signal operations that are completed with this semaphore which have a final value of 9. How many wait operations will take place?

☐ 11 operations

☐ 26 operations

☐ 38 operations

☐ 31 operations

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QUESTION: 48 GROUP: Computer Science SECTION: Operating System

Mark(s): 1

45

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A counting semaphore has a value of 31. It has 16 signal operations that are completed with this semaphore which have a final value of 9. How many semaphore?

☐ 11 operations

☐ 26 operations

☐ 38 operations

☐ 31 operations

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```
int main()
{
    int i = 0;
    for (i = 0; i < 20; i++)
    {
        switch(i)
        {
            case 0: i += 5;
            case 1: i += 2;
            case 5: i += 4;
            default: i += 4;
            break;
        }
        printf("%d ", i);
    }
    return 0;
}
```

☐ 5 9 13 17

☐ 15 20

☐ 12 17

☐ 0 5 9 13 17

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QUESTION: 45 GROUP: Computer Science SECTION: Operating System
Mark(s): 1

Which one of the given options is also called as the release time of a CPU scheduling process?

☐ Completion time

☐ Release time

☐ Start time

☐ Response time

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41

What will be the output of the program given below?

42 ★

```
#include <stdio.h>
```

43

```
#define ptr sell()
```

```
int a = 3, b = 4;
```

44 ☆

```
int main()
```

```
{
```

45

```
    a += (ptr & 2) << b - 1;
```

46

```
    do
```

47

```
    {
```

```
        a >>= b - 1;
```

48

```
    }while(1);
```

49

```
    printf("%d", a);
```

```
    return 0;
```

50

```
}
```

51

```
int sell()
```

52

```
{
```

```
    printf("%d", a += (b << a) ^ b - 1);
```

53

```
}
```

54

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QUESTION: 43 GROUP: Computer Science SECTION: OOPS
Mark(s): 1

Which of the feature given in options promotes access protection?

☐ Inheritance

☐ Composition

☐ Abstraction

☐ Encapsulation

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QUESTION: 50 GROUP: Computer Science SECTION: Operating System
Mark(s): 1

With how many priorities in a system a priority inversion problem will occurs?

☐ Only one priority

☐ Less than three priorities

☐ Only two priorities

☐ More than two priorities

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